

Zakk Lefkowitz

SENIOR SOFTWARE ENGINEER

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Experience

Amazon, Fire TV

Richmond, VA

SENIOR SOFTWARE ENGINEER

April 2020 - Present

Tech lead for Fire TV Catalog Enrichment, previously tech lead for Fire TV Partner Integrations. Promoted to Sr. Software Eng (Q4 2022).

Discovery Catalog Platform

Data Engineering, Distributed Systems

- Designed and built a governed data platform extending the Fire TV Catalog beyond streaming-provider feeds to enriched sources (Wikipedia articles, scene descriptions), replacing fragmented ad-hoc pulls by downstream teams.
- Drove the architectural decision to publish to a separate output stream rather than extending the core catalog, unlocking a **10x data-volume ceiling** and a faster schema-evolution path for novel data types.
- Aligned the Fire TV Search team to build its Prime Video scene-data curation app on the new platform, setting precedent for centralized data governance across downstream teams.
- Delivered to production solo in three months against a documented 4.5-month estimate. Enabled by agentic engineering.

LLM Genre Classification

LLM Applications & Evaluation, AWS Bedrock, Data Engineering

- Built an LLM genre classifier that replaced unreliable provider data, embedded in the catalog's stream-processing pipeline at **~\$0.001 per entity**; engineered an eligibility filter scoping inference to the ~5% of incoming updates to keep long-tail content off the latency-sensitive path.
- Invented **CMAR (Catalog Metadata Acceptance Rate)** - a measurement methodology using an LLM judge calibrated against PM ratings (>90% agreement on a 10K-entity sample) to validate enrichment quality. Classifier hit **95% CMAR vs. 82% baseline**; the metric earned PM confidence and unlocked downstream customer-team adoption of LLM-classified genre values.
- Tech lead on subsequent enrichment features (location extraction from linear-station metadata, image metadata extraction) shipped by mid-level engineers; core contributor to the catalog org's shared Bedrock integration library.

3P Continue Watching on Fire TV

Mobile-Cloud Architecture, Rust, Android, Java Services

- Led the client-side architecture on the VP-level initiative that gave Fire TV visibility into what customers watched inside partner apps, enabling Continue Watching and **8-figure incremental ad revenue**.
- Drove the decision to build the on-device middleware in Rust rather than piggyback on existing Android services. Shipped to 100M+ devices at <5MB memory footprint, proving out the native-service pattern subsequently adopted across Fire TV.
- Drove the decision to build a custom Amazon SDK rather than adopt Android's standard Watch Next API. Owned the entire partner-facing surface end-to-end (SDK, [integration docs](#), [reference sample app](#)); the resulting API surface became a partner-cited deal-closing factor across **Disney+, HBO Max, Hulu, Starz, and MGM+**.
- Built and operated the cloud ingestion backend handling sensitive customer and partner data, peaking at up to **~200K TPS**.
- Owned the privacy/consent surface end-to-end. Opt-in capture, consent handling, and implemented copy experimentation that drove **opt-in from 30% -> 75%** over a few months.

Agentic Engineering

Agentic Workflow Automation, MCP

- Authored and maintain **catalog-engineer** - a shared agent spec (context + MCP servers + skills) adopted 100% across a 3-team, ~40-person org, cited by VP as a best-practice example, and replicated by sister Fire TV orgs.
- Built and operate an internal **human-gated autonomous engineering pipeline**. Shipping production code through Amazon's standard peer code review. **Q1 2026 alone shipped ~70% as many merged PRs as all of 2025** (~3x output multiplier). Shared the system in an org-wide talk to 100+ engineers, driving 10+ direct adopters and two VP follow-ups.
- Trusted by Fire TV VP leadership to help provision a new AI Engineering velocity team; subsequently drove Catalog org adoption of a code package context graph system, landing **100 core packages** in 30 days.

CapTech Consulting

Richmond, VA

SENIOR CONSULTANT

July 2017 - March 2020

Led cloud modernization initiatives for Global Top 10 insurance client. Promoted to Sr. Consultant (Q3 2019).

Insurance Cloud Modernization

AWS ECS, CodeDeploy, Terraform, TypeScript, Angular, Node.js

- Architected **AWS ECS** migration and **CodeDeploy**-based CI/CD for a legacy Windows-based Java payment application processing ~\$1bn/year; built supporting **Terraform** IaC and a TypeScript deployment CLI.

Skills

Languages & Technologies

Rust, Java, Kotlin, TypeScript, Python, SQL, Android, Spring Boot, Apache Spark, RocksDB, MCP

AWS, Cloud & Infrastructure

EC2, ECS, Lambda, DynamoDB, SQS, Bedrock, CDK, S3, CloudWatch, CodeDeploy, Terraform (IaC)

Architecture & Core Competencies

Distributed Systems, Data Engineering, Mobile-Cloud Architecture, Agentic Workflow Automation, LLM Applications & Evaluation

Education

Virginia Polytechnic Institute and State University

Blacksburg, VA

B.S. COMPUTER SCIENCE, MATHEMATICS MINOR, CUM LAUDE

May 2017